Project Hello World

Hello Hub Lite sample

Community-powered connectivity planning

Hello Solar Planner
Planning estimate
May 23, 2026

Executive Summary

Total daily energy **4.84 kWh**	**Peak load** **405 W**	**Surge load** **438 W**	**Critical energy** **4.23 kWh**

This estimate is prepared for the selected Fully DC System planning path. Other planner options remain available in the app for comparison.

Project Summary

Country **Uganda**	**Currency** **USD**	**USD exchange rate** **1 USD = 1 USD**
System voltage **24 V**	**Sun hours** **4.8 h/day**	**Autonomy** **1.5 day(s)**
Report option **Fully DC System**	**Brand profile** **Project Hello World**	

Load Table

LOAD	QTY	W EACH	H/DAY	TYPE	VOLTAGE	SURGE	CRITICAL	DAILY WH
Mikrotik Router	5	18	24	DC	24 V	1.2x	Yes	2,160
Outdoor access point	6	12	24	DC	24 V	1.2x	Yes	1,728
Learning tablets	10	8	3	DC	5 V	1x	No	240
LED lights	6	5	5	DC	12 V	1.1x	Yes	150
USB charging station	1	60	4	DC	12 V	1.1x	No	240
Monitoring gateway	1	8	24	DC	12 V	1.2x	Yes	192
Technician laptop charging	1	65	2	AC	230 V	1.5x	No	130
Total daily energy								**4,840 Wh**

Technical Sizing Summary

METRIC	FULLY DC SYSTEM
Adjusted daily energy	5.15 kWh
Required LiFePO4 battery	11.2 kWh
Recommended solar array	1,650 W
Recommended MPPT current	90 A
Recommended inverter size	Not required

Generated / Edited Equipment Plan

EQUIPMENT	CURRENT PLAN	ACTUAL CAPACITY	NOTES
Solar panels	6 panel(s) x 360 W	2,160 W	Requirement: 1,650 W
LiFePO4 batteries	5 battery/batteries x 24 V x 100 Ah	12 kWh	Requirement: 11.2 kWh
Fully DC MPPT/controller	1 controller(s) x 90 A	90 A	Requirement: 90 A

Fully DC System

Status: Pass

Best when AC devices can be replaced with DC equivalents or powered through small point-of-use adapters.

CATEGORY	RECOMMENDATION	RATIONALE
Energy target	5.1 kWh adjusted daily demand	Uses DC distribution efficiency to account for wiring, conversion, and operating margin.
Battery	11.2 kWh LiFePO4 minimum usable-backed storage	Sized for 1.5 autonomy day(s) at the configured depth of discharge.
Solar array	1,650 W PV array	Based on 4.8 average sun hour(s) with derating for real-world conditions.
Charge control	90 A MPPT at 24 V	Controller current includes a safety factor above expected PV charging current.

Financial Summary

Fully DC System option **USD 6,718**

Fully DC System Cost Detail

CATEGORY	DESCRIPTION	QTY	UNIT COST	TOTAL
Solar panels	6 panel(s) x 360 W	6	USD 245 (USD 245)	USD 1,470
LiFePO4 battery storage	5 battery/batteries x 24 V x 100 Ah	5	USD 560 (USD 560)	USD 2,800
Charge controller or hybrid inverter	1 controller(s) x 90 A MPPT	1	USD 260 (USD 260)	USD 260
DC distribution and protection	DC breaker board, fuses, labels, and surge protection	1	USD 180 (USD 180)	USD 180
Cabling and connectors	PV, battery, and load cabling with MC4/connectors	1	USD 210 (USD 210)	USD 210
Earthing and lightning protection	Earthing rod, bonding, surge, and lightning protection kit	1	USD 190 (USD 190)	USD 190
Monitoring	Battery monitor and remote energy logging	1	USD 155 (USD 155)	USD 155
Installation	Planning allowance based on editable assumption	1	16%	USD 842
Contingency	Planning allowance for local variance and missing items	1	10%	USD 611
Estimated total				**USD 6,718 (USD 6,718)**

Assumptions

DC distribution efficiency	**94%**	**Hybrid DC efficiency**	**94%**	**Inverter efficiency**	**88%**
Battery depth of discharge	**80%**	**Battery reserve factor**	**115%**	**PV derate factor**	**75%**
MPPT safety factor	**125%**	**Inverter headroom factor**	**125%**	**Panel price**	**USD 245 / panel**
Battery price	**USD 560 / battery**	**Inverter price**	**USD 690 / inverter**		

Prices are editable planning assumptions and should be localized before procurement.

Safety Disclaimer

This report is for planning estimates only, not certified electrical design. Final installation must be reviewed by a qualified solar/electrical technician and comply with local electrical, structural, grounding, and lightning protection requirements.

Prepared for Project Hello World community connectivity planning.